

Weiwu Yan

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Education

University of Chicago	Fall 2023 – Spring 2025
• Master of Arts in Computational Social Science	
University of California, San Diego	Fall 2019 – Spring 2023
• Bachelor of Science in Social Psychology • Bachelor of Arts in Sociology	

Projects

Gender and Emotional Expression in Video Game Dialogue	Spring 2025
• Collected and processed 247,644 dialogue lines from 3,043 characters across 18 role-playing video games from the Video Game Dialogue Corpus. • Conducted emotion classification using RoBERTa (Hugging Face) to analyze gender differences in emotional expression. • Performed Part-of-Speech (spaCy) tagging and TF-IDF vectorization (scikit-learn) to identify the verbs most uniquely associated with each gender. • Utilized regular expressions to examine gender differences in the distribution of terminal punctuation.	
A Real-Time Chat Application	Fall 2024
• Designed the frontend of a chat application (React) with features such as a navigation bar, interactive hover effects, and responsive layout adaptation. • Developed the backend of a chat application (Flask) with features such as user account management, channel-based communication, and real-time messaging for text, emojis, and image sharing.	
Multiclass Classification of Scholarly Articles	Spring 2024
• Collected and processed metadata of 1.7 million scholarly articles from the arXiv Dataset. • Performed Multinomial Logistic Regression (PySpark) on AWS to identify latent topics across articles. • Trained a Latent Dirichlet Allocation model (PySpark) on AWS to classify articles into research fields.	
An Agent-Based Model of Urban Gentrification	Spring 2024
• Designed an agent-based model (Mesa) to simulate urban gentrification dynamics, examining how citizens and immigrants with varying socioeconomic status interact and relocate within a city.	
Funding and Academic Output in Behavioral and Cognitive Sciences	Winter 2023
• Collected and processed 4,959 funding records from U.S. NSF Division of Behavioral and Cognitive Sciences. • Scraped metadata of 207,370 scholarly articles authored by 3,230 identified researchers from Google Scholar. • Applied k-means clustering and TF-IDF vectorization (scikit-learn) to examine the relationship between seven identified fields and funding amounts. • Utilized a Structural Topic Model (R) to analyze temporal trends across 12 identified topics. • Performed document embeddings using SciBERT (Hugging Face) to assess the impact of funding on research diversity.	
Multiclass Classification of Video Games	Winter 2023
• Collected and processed metadata of 75,776 video games from the Steam Games Dataset. • Trained a Random Forest Classifier (scikit-learn) to classify games based on textual descriptions. • Trained a Convolutional Neural Network (TensorFlow) to classify games based on cover art.	

Lineup Procedure and Confidence in Eyewitness Identification

Summer 2022

- Collected and processed responses from 78 participants in a between-subjects factorial design experiment (Qualtrics).
- Conducted an ANOVA test (Jamovi) to analyze the effects of lineup procedure and confidence on eyewitness identification accuracy.

Moral Judgment of Maternal Occupation and Responsibility

Fall 2020

- Collected and processed responses from 285 participants in an observational study (Qualtrics).
- Conducted an ANOVA test (Jamovi) to analyze the relationship between maternal occupation and moral judgment of mothers leaving their children unsupervised.

Skills

Languages: Chinese • English

Programming: Python • SQL • JavaScript • R

Machine Learning: Hugging Face • TensorFlow • PyTorch • scikit-learn • spaCy • Gensim

Data Analysis: Pandas • NumPy • SciPy • dplyr • Jamovi

Data Visualization: Plotly • ggplot2 • Tableau • Matplotlib • Mesa

High-Performance Computing: Amazon Web Services • PySpark

Web Development: Shiny • Flask • React

Survey: Qualtrics • Google Survey